FM BROADCAST SYSTEM

CO2 Engineering Case Study – Design Sheet

Objective: design and evaluate a regional FM broadcast chain using the supplied requirements: 15 kHz broadcast audio, 100 MHz candidate carrier, 75 kHz initial peak deviation, AWGN/practical interference, limiter-discriminator receiver and MATLAB verification. The supplied case study requires a complete requirement-to-deployment engineering process. □filecite□turn3file0□L19-L48□

Parameter	Requirement	Design basis
Audio	15 kHz max	0–15 kHz
Carrier	100 MHz	100 MHz
Deviation	75 kHz initial	75 kHz
Channel	AWGN + interference	Noise + selectivity
Receiver	Limiter + discriminator	Superheterodyne
Noise improvement	Pre/de-emphasis	75 μs analytical model
Platform	MATLAB	Waveforms + spectra + AWGN

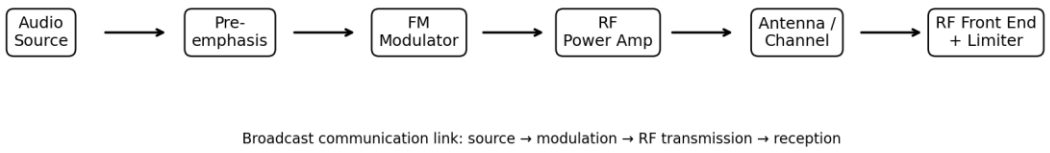


Figure 1. Overall broadcast communication chain.

1. REQUIREMENT INTERPRETATION

The system must provide reliable audio while controlling occupied spectrum, RF power, receiver complexity and noise. The assignment asks for explicit assumptions, system constraints and a final operating basis.
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Design item	Selected basis
Message bandwidth	15 kHz
Carrier	100 MHz
Initial Δf	75 kHz
Maximum β	5
Noise model	AWGN; practical interference discussed
Detection	Limiter + discriminator
Noise improvement	Pre-emphasis/de-emphasis
Verification	MATLAB

Decision 1: adopt the 100 MHz WBFM operating basis. The 75 kHz deviation is retained initially because it is the specified starting point; final acceptance depends on bandwidth, noise and channel-mask verification.

2. ANGLE MODULATION: FM VS PM

FM: $s(t)=A_c \cos[2\pi f_c t + 2\pi k_f \int m(\tau)d\tau]$

PM: $s(t)=A_c \cos[2\pi f_c t + k_p m(t)]$

Criterion	FM	PM
Information	Frequency variation	Phase variation
Index	$\beta=\Delta f/fm$	Phase index
Detection	Discriminator/PLL	Phase detector/PLL
Noise	Excellent amplitude-noise rejection	Constant-envelope benefit
Broadcast fit	Directly matches FM receiver	Less direct for this case

$$\beta = \Delta f/f_m = 75/15 = 5$$

$\beta = 5$ is wideband FM. FM is selected because the case explicitly provides a frequency deviation and a frequency-discriminator receiver, and because its constant-envelope property is valuable in practical RF transmission. The assignment requires quantitative FM/PM comparison and a defended selection. □filecite□turn3file0□L71-L86□

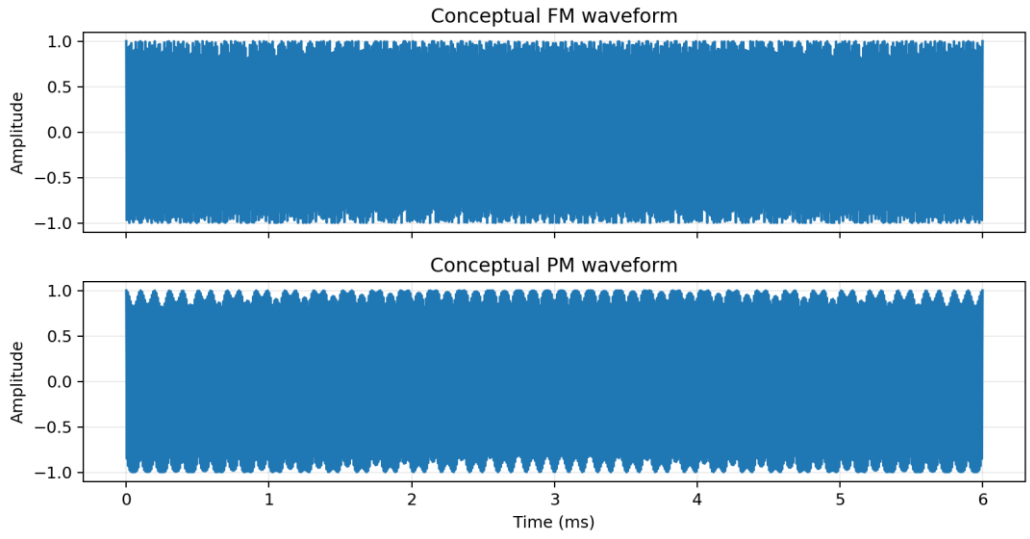


Figure 2. Conceptual FM and PM waveforms.

3. FM MODULATION INDEX & BASIC FREQUENCY LIMITS

$\beta_{max} = 75\ kHz / 15\ kHz = 5$

$f_{inst,max} = 100\ MHz + 75\ kHz = 100.075\ MHz$

$f_{inst,min} = 100\ MHz - 75\ kHz = 99.925\ MHz$

Quantity	Value
Carrier	100 MHz
Maximum audio frequency	15 kHz
Peak deviation	75 kHz
Modulation index	5
Frequency swing	±75 kHz
FM type	WBFM

Increasing deviation raises the modulation index and generally improves high-C/N FM noise performance, but also increases occupied spectrum and adjacent-channel risk. Reducing deviation has the opposite trade-off.

4. FM SPECTRUM & BESSEL ANALYSIS

For a single-tone FM signal, the carrier and sidebands occur at $f_c \pm n f_m$. Their amplitudes are controlled by Bessel functions $J_n(\beta)$. For $\beta = 5$, several sideband pairs are significant.

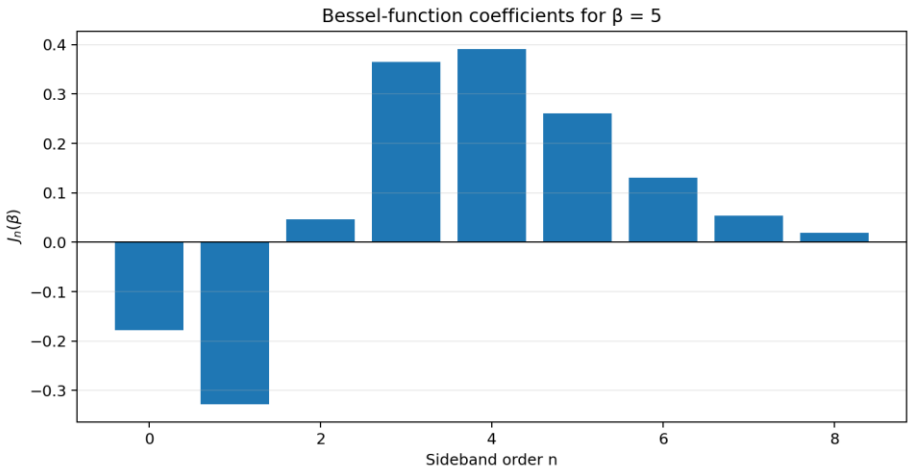


Figure 3. Bessel coefficients J_n(5).

Order	Approx. coefficient	Observation
0	−0.178	Carrier component
1	−0.328	Strong
3	+0.365	Strong
4	+0.391	Strong
5	+0.261	Significant
6	+0.131	Visible
7	+0.053	Smaller
8	+0.018	Near practical cutoff

The infinite mathematical spectrum is truncated in practice by a chosen significance criterion. This is why a Bessel-sideband estimate can differ from Carson's rule.

5. FM TRANSMISSION BANDWIDTH

$B_{Carson} \approx 2(\Delta f + f_{m,max})$

$B_{Carson} = 2(75 + 15) = 180$ kHz

Approximate Carson edges: 99.910 MHz to 100.090 MHz

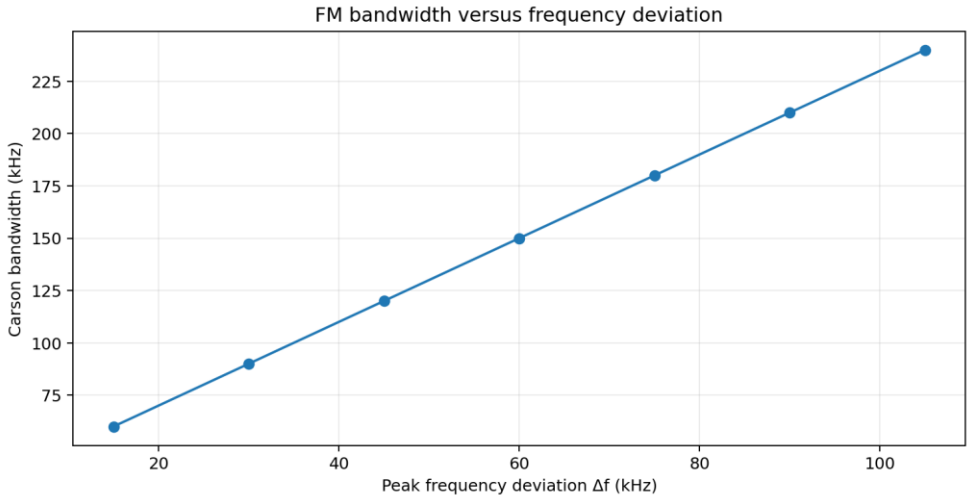


Figure 4. Carson bandwidth versus peak frequency deviation.

Δf	β at 15 kHz	Carson BW
45 kHz	3	120 kHz
75 kHz	5	180 kHz
90 kHz	6	210 kHz

Decision 2: retain 75 kHz as the baseline because it gives β = 5 and 180 kHz Carson bandwidth, balancing audio/noise performance against spectrum occupancy. The supplied rubric specifically requires Carson's rule, significant-sideband comparison and adjacent-channel implications. □filecite□turn3file0□L88-L108□

6. FM TRANSMITTER DESIGN

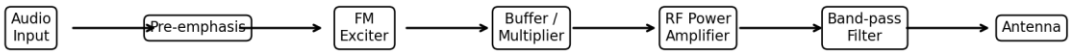


Figure 5. Proposed FM transmitter architecture.

Block	Function
Audio input	Programme audio, 0–15 kHz
Pre-emphasis	Boost high-frequency audio
FM exciter	Creates controlled frequency deviation
RF buffer/multiplier	Isolation/frequency conversion
RF power amplifier	Raises transmitted RF power
Band-pass filter	Suppresses harmonics/spurious emissions
Antenna	Radiates 100 MHz signal

Preferred implementation: a stable FM exciter followed by RF amplification and filtering. Direct or indirect FM generation may be used, but the implementation must control carrier frequency and deviation accurately.

7. FM RECEIVER DESIGN

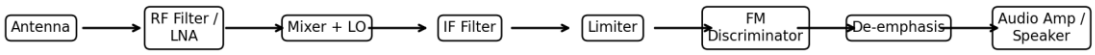


Figure 6. Proposed superheterodyne FM receiver.

Stage	Purpose
RF filter/LNA	Select station; improve sensitivity
Mixer + LO	Translate RF to fixed IF
IF filter	Select channel; reject adjacent stations
Limiter	Remove amplitude variations/noise
Discriminator	Convert frequency variation to audio voltage
De-emphasis	Restore audio response; reduce high-frequency noise
Audio amplifier	Drive speaker/load

Decision 3: use a superheterodyne architecture with limiter and discriminator. The supplied case requires these receiver functions and asks for sensitivity, selectivity, fidelity, stability and adjacent-channel rejection analysis. □filecite□turn3file0□L110-L126□

8. IF SELECTION & IMAGE-FREQUENCY ANALYSIS

Choose f_IF = 10.7 MHz for the VHF FM receiver.

f_LO = f_RF + f_IF = 100 + 10.7 = 110.7 MHz

f_image = f_RF + 2f_IF = 100 + 21.4 = 121.4 MHz

Parameter	Value
RF carrier	100 MHz
IF	10.7 MHz
High-side LO	110.7 MHz
Image frequency	121.4 MHz
Detection	Limiter + discriminator

The RF front end must reject the image and strong adjacent stations before mixing. The fixed IF then provides predictable filtering and selectivity.

Criterion	TRF	Superheterodyne
Selectivity	Varies with tuning	Strong fixed-IF filtering
Sensitivity	More difficult over tuning range	Good practical performance
Complexity	Lower	Higher
Broadcast suitability	Limited	Preferred

9. AM & DSB-SC NOISE BENCHMARK

AM and DSB-SC are required reference systems. AM transmits a carrier plus sidebands; DSB-SC removes the carrier and therefore improves carrier-power utilization but requires coherent detection. □filecite□turn3file0□L128-L141□

AM total power: $P_T = P_c(1 + \mu^2/2)$

Feature	AM	DSB-SC
Carrier	Present	Suppressed
Detector	Envelope	Coherent
Carrier synchronization	Not essential	Essential
Power efficiency	Lower	Higher
Receiver complexity	Low	Higher

For noise analysis, model n(t) as zero-mean Gaussian white noise with PSD $N_0/2$ over the relevant receiver bandwidth. Use the same signal/noise reference bandwidth when comparing SNR. DSB-SC is a useful benchmark but is less convenient for conventional broadcast because carrier recovery is required.

10. FM NOISE & THRESHOLD EFFECT

$(S/N)_{audio} \propto (3/2)\,\beta^2\,(C/N)$ at sufficiently high C/N

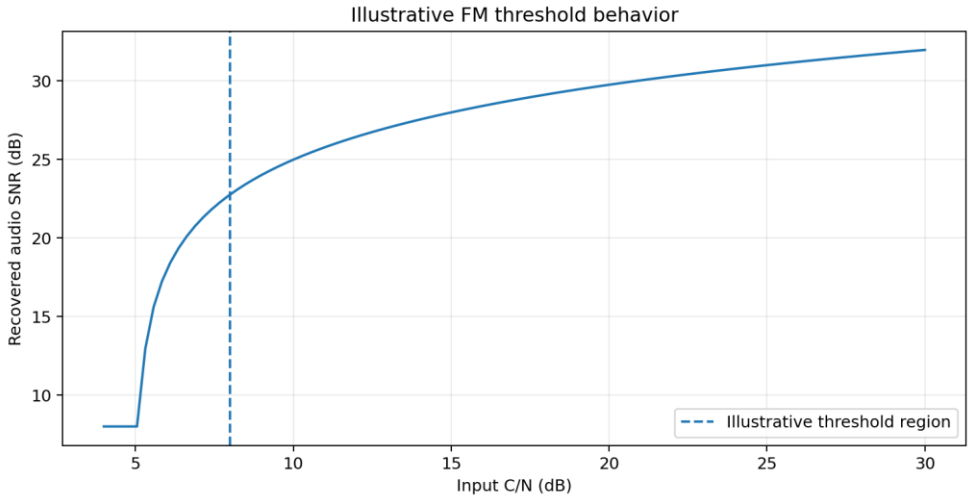


Figure 7. Illustrative FM threshold behaviour.

C/N region	Expected result
High	Clean recovered audio
Moderate	Increasing background noise
Near threshold	Rapid quality loss
Below threshold	Severe degradation

The limiter suppresses amplitude fluctuations before the discriminator. At low C/N, however, the FM detector enters threshold behaviour and recovered audio quality can collapse rapidly. The assignment requires progressive C/N reduction and evidence of threshold behaviour. □filecite□turn3file0□L143-L166□

Initial operating recommendation: maintain a practical C/N margin comfortably above the measured threshold; about 20 dB is a reasonable simulation target, subject to the actual MATLAB results.

11. PRE-EMPHASIS & DE-EMPHASIS

$H_{pre}(s)=1+s\tau$; $H_{de}(s)=1/(1+s\tau)$

For $\tau = 75\ \mu s$: $f_c = 1/(2\pi\tau) \approx 2.12$ kHz

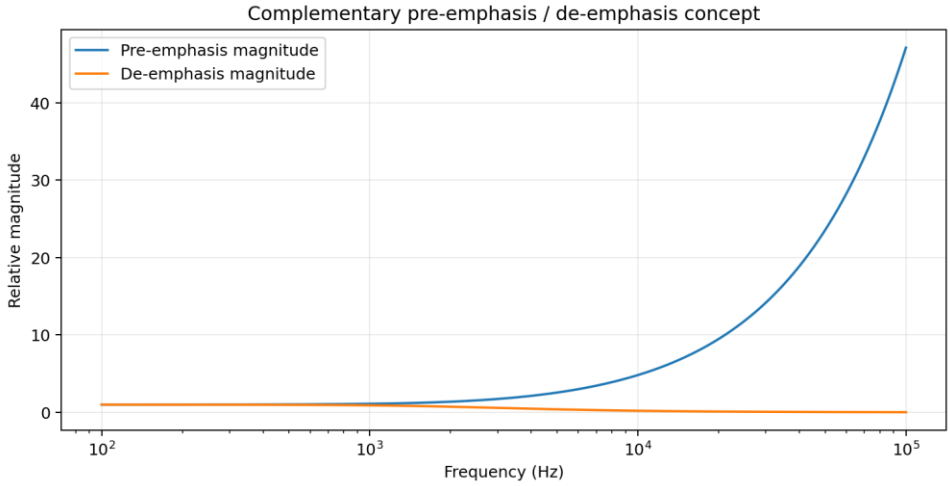


Figure 8. Complementary pre-emphasis/de-emphasis response.

Aspect	Without pair	With pair
High-frequency noise	Higher	Reduced after de-emphasis
Audio response	Direct	Complementary restoration
Complexity	Lowest	Small added stage
Risk	Less processing	Avoid pre-emphasis clipping

Decision 4: use complementary pre-emphasis/de-emphasis. The transmitter must control audio level so the boosted high-frequency content does not cause excessive deviation or clipping. The supplied assignment explicitly requires a with/without comparison and a bandwidth/noise/audio-response trade-off. □filecite□turn3file0□L154-L166□

12. MATLAB VERIFICATION PLAN

Test	MATLAB task	Expected evidence
1	Generate 15 kHz message	Correct message band
2	Generate FM and PM	Angle-modulated waveforms
3	Measure β	$\beta \approx 5$
4	FFT	Carrier + Bessel sidebands
5	Bandwidth	Near 180 kHz Carson estimate
6	Add AWGN	Controlled C/N sweep
7	FM demodulation	Recovered audio
8	Threshold sweep	Rapid SNR degradation
9	Pre/de-emphasis	Improved HF noise
10	AM/DSB-SC	Reference SNR comparison

MATLAB results must be interpreted, not simply pasted. Explain finite record length, FFT leakage, windowing, filter roll-off and the selected occupied-bandwidth criterion whenever theory and simulation differ.

The supplied Stage 7 requires message/FM/PM waveforms, spectra, bandwidth verification, AWGN at multiple C/N levels, recovered audio, threshold plots and pre/de-emphasis comparison. □filecite□turn3file0□L170-L190□

13. FINAL SPECIFICATION–VERIFICATION

Requirement	Specified	Achieved	Status
Message bandwidth	15 kHz	15 kHz	YES
Carrier	100 MHz	100 MHz	YES
Peak deviation	75 kHz	75 kHz	YES
FM index	Calculated	$\beta=5$	YES
FM bandwidth	Calculated	180 kHz	YES*
Noise	AWGN/interference	Modelled	YES*
Pre/de-emphasis	Selected	75 μs model	YES*
Receiver	Selected	Superhet + 10.7 MHz IF	YES
MATLAB	Required	Verification plan	PENDING RUN

*Final status depends on actual MATLAB measurements and the applicable broadcast regulations.

Submission flow: REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → DESIGN → SIMULATION → VERIFICATION → DEPLOYMENT DECISION

14. FINAL ENGINEERING DECISION

DEPLOYMENT RECOMMENDATION

FINAL DECISION: YES — DEPLOY THE BASELINE FM ARCHITECTURE AFTER FINAL MATLAB AND REGULATORY VERIFICATION.

Decision area	Final recommendation
Modulation	Wideband FM
Carrier	100 MHz
Audio bandwidth	15 kHz
Peak deviation	75 kHz
Modulation index	$\beta = 5$
Carson bandwidth	180 kHz
Transmitter	Pre-emphasis → FM exciter → RF PA → filter
Receiver	Superheterodyne → limiter → discriminator → de-emphasis
IF	10.7 MHz
Image	121.4 MHz
Noise strategy	Pre/de-emphasis + adequate C/N margin

Engineering justification:

- FM provides constant-envelope transmission and strong amplitude-noise rejection.
- 75 kHz deviation gives $\beta = 5$ and a practical 180 kHz Carson bandwidth.
- Superheterodyne reception provides predictable IF selectivity and sensitivity.
- Limiter/discriminator detection matches the supplied receiver requirement.
- Pre-emphasis/de-emphasis improves high-frequency recovered-audio noise performance.
- Final deployment must verify the actual C/N threshold, measured occupied bandwidth, emission mask, frequency accuracy and local regulatory requirements.

If the MATLAB study shows inadequate threshold margin, first improve received C/N through antenna/feed-loss optimization, RF filtering, receiver sensitivity and coverage. Do not increase deviation without checking the resulting spectrum and channel compliance. The assignment requires a clear YES/NO decision defended using modulation, bandwidth, architecture, SNR, threshold and pre/de-emphasis evidence. □filecite□turn3file0□L192-L205□

END OF ENGINEERING CASE STUDY DESIGN SHEET